

QUESTIONS 1/5 completed

Hash-Linear Probing

HASH-LINEAR PROBING

Write a program to create a hash table of size 10, insert the elements 25, 35, 15, 45, 95. If any collisions occurred, then resolve collisions using linear probing. Finally display the hash table after each insertion.

PROGRAM EDITOR

C

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```
1 #include <stdio.h>
2
3 int main() {
4     int table[10];
5     int size = 10;
6     int i, j;
7     int elements[] = {25, 35, 15, 45, 95};
8     int num_elements = 5;
9
10    for (i = 0; i < size; i++) {
11        table[i] = -1;
12    }
13
14    for (i = 0; i < num_elements; i++) {
15        int key = elements[i];
16        int index = key % size;
17
18        while (table[index] != -1) {
19            index = (index + 1) % size;
20        }
21    }
```

OUTPUT

```
After inserting 25: - - - - - 25 - - - -
After inserting 35: - - - - - 25 35 - - - -
After inserting 15: - - - - - 25 35 15 - - -
After inserting 45: - - - - - 25 35 15 45 -
After inserting 95: - - - - - 25 35 15 45 95
```

INPUT

```
10
25 35 15 45 95
```

Continue

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Hash-Linear Probin

HASH-LINEAR PROBING

Write a program to create a hash table of size 10, insert the elements 25, 35, 15, 45, 95. If any collisions occurred, then resolve collisions using linear probing. Finally display the hash table after each insertion.

```
16 int index = key % size;
17
18 while (table[index] != -1) {
19     index = (index + 1) % size;
20 }
21
22 table[index] = key;
23
24 printf("After inserting %d: ", key);
25 for (j = 0; j < size; j++) {
26     if (table[j] == -1) {
27         printf("- ");
28     } else {
29         printf("%d ", table[j]);
30     }
31 }
32 printf("\n");
33 }
34
35 return 0;
36 }
```

OUTPUT

```
After inserting 25: - - - - 25 - - - -
After inserting 35: - - - - 25 35 - - -
After inserting 15: - - - - 25 35 15 - -
After inserting 45: - - - - 25 35 15 45 -
After inserting 95: - - - - 25 35 15 45 95
```

10

25 35 15 45 95